

γ (beam π^0) — same architecture, with and without the provenance features

Model	Efficiency (%)	Purity (%)	Purity × Efficiency (%)	Threshold
Cut-based	48.3	69.5 ± 0.4	33.56	-
Hybrid base4	48.3	69.0 ± 0.4	33.32	0.661
Hybrid all8	48.3	75.2 ± 0.4	36.30	0.683